SSDiffNet: Semantic-Guided and Structure-Consistent Diffusion for Aerial Visible-to-Infrared Image Translation

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***Abstract*—Visible-to-infrared image translation in aerial scenarios is fundamentally challenged by severe cross-modal appearance discrepancies, complex land-cover semantics, and large variations in brightness distribution. Existing methods, largely developed for natural scenes, often fail to jointly preserve semantic correspondence and fine-grained structural consistency when applied to aerial imagery, leading to semantic misalignment and local structural distortion despite visually plausible global appearance. To address these challenges, we propose SSDiffNet, a diffusion-based framework that explicitly enforces semantic alignment and structural consistency for aerial visible-to-infrared image translation. Instead of relying on global appearance matching, SSDiffNet guides the diffusion process with cross-modal semantic priors while constraining structural stability throughout generation. Specifically, an infrared image adaptive module (IIAM) adapts the diffusion decoder to infrared imaging characteristics under joint spatial- and frequency-domain constraints, enabling more faithful infrared-style reconstruction. A cross-modal semantic feature guidance module (CSFGM) injects learned visible-to-infrared semantic priors into multilevel denoising features, effectively enhancing regional semantic correspondence. In addition, a dual structural consistency constraint module (DSCCM) enforces invariant structural representations under equivalent transformations, substantially reducing structural deviation and local distortion during generation. Extensive experiments conducted on three public aerial visible-to-infrared datasets demonstrate that SSDiffNet consistently outperforms existing state-of-the-art methods in both visual quality and quantitative metrics.**

***Index Terms*—Aerial visible-to-infrared image translation, cross-modal semantic alignment, diffusion models.**

I. Introduction

INFRARED images play a critical role in aerial perception tasks such as military reconnaissance, fire monitoring, concealed target detection, and surface thermal activity analysis, owing to their robustness under low light conditions, occlusion, and complex weather environments [1], [2], [3], [4], [5], [6]. Infrared and spectral cues have also shown effectiveness in bad-weather perception and remote sensing anomaly analysis [7], [8]. Compared with visible imaging, infrared imaging is inherently less sensitive to illumination variations [9]. However, the high cost of infrared sensors, restrictive deployment conditions, and strong dependence on environmental and sensor factors severely limit the availability of high-quality infrared data [6], posing a major challenge to the development of aerial infrared-based applications.

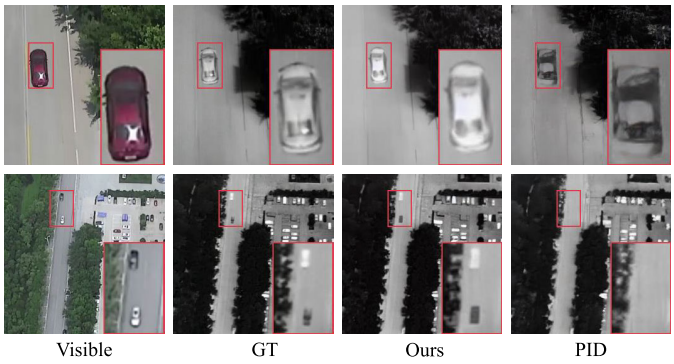


Fig. 1. Visual comparison with PID. Compared with our method, PID still shows deficiencies in local details and noticeable deviations in semantic attributes such as brightness.

Visible-to-infrared image translation has emerged as a promising solution to alleviate infrared data scarcity by generating infrared images from easily accessible visible images [2], [10], [11], [12], [13], [14]. Nevertheless, visible-to-infrared translation is fundamentally challenging due to substantial cross-modal discrepancies in imaging mechanisms, radiation characteristics, and brightness distributions, making the task substantially more challenging than simple appearance transfer [15]. In aerial scenarios, complex spatial structures, multi-scale targets, and diverse backgrounds in thermal radiation further exacerbate these challenges, often resulting in semantic misalignment and structural distortion in generated infrared images [16], [17], as illustrated in Fig. 1.

To improve generation quality, existing studies explore visible-to-infrared translation from different perspectives. Some methods incorporate constraints into generative adversarial frameworks to enhance stability [18], [19], [20]. More recently, diffusion models have been introduced into infrared image generation due to their training stability and strong generative capability, modeling the noise evolution process in latent space to improve overall image quality [15]. In parallel,

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recent studies on multimodal remote sensing have explored complementary information modeling through multimodal representation learning [21], global–local feature modeling [22], intermodal and cross-modal learning [23], and disentangled representation learning for cross-domain adaptation [24].

Despite these advances, most existing methods still rely on indirect semantic or structural cues extracted solely from visible images [25], [26], [27] and fail to establish explicit and stable cross-modal semantic correspondence between visible and infrared images. Moreover, current latent diffusion approaches typically adopt variational autoencoders (VAEs) trained on natural visible images as latent priors [28], [29], neglecting the substantial differences in texture distribution, brightness structure, and noise characteristics of infrared imagery. These limitations hinder the preservation of infrared-specific thermal patterns and structural details, leading to semantic mismatch and structural deviation in generated results.

To address these challenges, we propose SSDiffNet, a diffusion-based framework for aerial visible-to-infrared image translation that emphasizes semantic alignment and structural consistency. To reduce the distribution mismatch between visible and infrared images, an infrared image adaptive module (IIAM) is designed to tailor latent reconstruction toward infrared imaging characteristics. To explicitly model cross-modal semantic correspondence and mitigate semantic drift, a cross-modal semantic feature guidance module (CSFGM) provides semantic guidance during the diffusion denoising process, promoting consistent regional alignment across modalities. Furthermore, to suppress structural deviation in complex aerial scenes, a dual structural consistency constraint module (DSCCM) enforces structural stability under geometric transformations, enhancing the robustness of generated infrared images. Together, these components enable SSDiffNet to achieve stable and reliable visible-to-infrared translation while preserving both semantic coherence and structural integrity. The main contributions of this work are summarized as follows.

1) We propose SSDiffNet, a diffusion-based framework for aerial visible-to-infrared image translation that explicitly enforces semantic alignment and structural consistency, achieving stable and reliable generation in complex aerial scenarios.
2) We design a cross-modal modeling strategy that integrates infrared adaptive latent reconstruction and semantic-guided diffusion denoising, enabling explicit visible-to-infrared semantic correspondence and effectively reducing semantic drift during translation.
3) We introduce a dual structural consistency constraint that enforces invariant structural representations under geometric transformations, significantly improving structural fidelity. Extensive experiments on multiple aerial visible-to-infrared datasets demonstrate consistent performance gains over state-of-the-art methods.

The rest of this article is organized as follows. Section II reviews the related work, Section III presents the proposed method, and Section IV reports the experimental results and analyses. Section V discusses the findings and limitations, and Section VI concludes this article.

II. Related Work

A. Diffusion Models

In recent years, diffusion models have gradually become an important research direction in the field of image generation [30], [31], [32] and have demonstrated strong potential in terms of generation quality and training stability. These methods introduce noise into the data distribution during the training stage and learn the reverse denoising process to progressively recover clean samples, thereby enabling effective modeling of complex image distributions. Compared with traditional generative models such as GANs [33] and VAEs [34], diffusion models usually achieve more stable optimization behavior in high-resolution image generation tasks [35], [36], [37] and alleviate issues such as mode collapse. Early studies systematically established diffusion frameworks based on denoising probabilistic modeling, which laid the theoretical foundation for subsequent research. Following this, extensive efforts have focused on model architecture design [38] and sampling efficiency [39], [40], leading to the widespread application of diffusion models in tasks such as image synthesis [41], [42], editing [43], [44], and restoration [45], [46].

To reduce the computational cost caused by diffusion processes in high-dimensional pixel space, some studies transfer diffusion modeling to compressed latent spaces [28], which significantly improves computational efficiency. However, the application of diffusion models in multispectral or cross-modal image translation tasks remains relatively limited.

B. Image-to-Image Translation

Image translation methods focus on establishing cross-domain mappings by modeling either explicit pixel-level relationships or implicit distribution-level representations. For visible-to-infrared image translation, existing studies explore multistyle modeling, physical priors, and guidance from auxiliary information. PID [15] integrates infrared imaging mechanisms into diffusion frameworks, while CM-Diff [25] achieves bidirectional visible-to-infrared generation through a unified diffusion framework. DiffV2IR [26] and F-ViTA [27] leverage vision–language information for improved scene understanding.

For diffusion-based image translation, BBDM [29] models the diffusion path from source to target images using a Brownian bridge mechanism, enabling image-to-image translation.

For GAN-based image translation, Isola et al. [47] proposed Pix2Pix with a U-Net generator and a PatchGAN discriminator, enabling end-to-end conditional image translation and forming the basis of many subsequent methods. CycleGAN [48] and DiscoGAN [49] maintain cross-domain correspondence through bidirectional generation. Subsequent studies have further shifted from pixel-level constraints to latent and feature space modeling, including joint latent variable modeling [50], content-style disentanglement [51], domain-specific content mapping [52], and shared content attribute space modeling [53], with distribution-level adversarial consistency

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used to improve flexibility and diversity [54]. Swin-UNIT [55], ITTR [56], and UVCGAN [57] incorporate transformer architectures for enhanced feature extraction.

DreamBooth [58] introduced textual prompts to enable targeted image modification. To address the difficulty of preserving content consistency in single-image editing with diffusion models, Zhang et al. [59] proposed SINE to enhance structural stability, while Wang et al. [60] improved cross-domain generation efficiency and stability by replacing the commonly used discrete latent space with a binary latent space. InstructPix2Pix [61], ControlNet [62], and T2I-Adapter [63] achieve controllable generation under multimodal constraints through instruction data or conditional branches, with related prompt designs also explored in spectral modeling [64]. Although these methods show promising progress, aerial visible-to-infrared translation remains challenging due to complex scene semantics and structural variations, making direct transfer of existing approaches difficult. Therefore, we focus on aerial visible-to-infrared translation, aiming to generate reliable infrared images from aerial visible inputs.

C. Aerial Visible-to-Infrared Image Translation

Building upon general image-to-image translation frameworks, aerial visible-to-infrared image translation methods have recently attracted increasing attention in remote sensing. Han et al. [2] provided a comprehensive survey of this research field. Li et al. [16] enhance the expression of infrared radiation characteristics through thermal-feature-guided visible-to-infrared translation. USTNet [17] improves the modeling capability of multiscale structures and long-range dependencies by integrating a Swin Transformer. Ma et al. [19] jointly model cross-modal generation and template matching tasks to enhance the task usability of the generated results, while template update strategies have been investigated in related work [65]. Ma et al. [66] introduce contrastive constraints to further separate style information between visible and infrared images. DR-AVIT [67] adopts a content-style disentanglement strategy to enhance generation diversity, while Zhao et al. [68] explicitly model the temperature distributions of objects and backgrounds to achieve more controllable and diverse infrared image generation. Spectral adaptive aggregation provides another strategy for modeling spectral cues [69].

Although these methods introduce targeted designs for task usability, style diversity, or infrared radiation modeling, most of them still lack explicit guidance for high-level semantic relationships during visible-to-infrared mapping. In addition, structural consistency under equivalent geometric transformations is not fully explored. As a result, semantic drift and loss of fine structural details may occur in complex aerial scenarios. To address these issues, we introduce cross-modal semantic guidance and dual structural consistency mechanisms to improve the stability and consistency of infrared image generation.

III. Proposed Method

A. Overall Architecture

To overcome the challenges of semantic misalignment and structural distortion in aerial visible-to-infrared image translation, we propose SSDiffNet, a diffusion-based framework that integrates infrared adaptive modeling, cross-modal semantic guidance, and structural consistency constraints to produce reliable and faithful infrared images. As shown in Fig. 2, the framework consists of three key components: an IIAM, a CSFGM, and a DSCCM. The IIAM introduces a spatial–frequency collaborative reconstruction mechanism to adapt the generation process toward infrared characteristics. The CSFGM learns cross-modal semantic consistency via contrastive learning and injects semantic features for semantic alignment. Finally, the DSCCM employs a dual-branch structure with shared weights, enforcing structural and semantic consistency in the generated infrared images.

B. Infrared Image Adaptive Module

To better align the generated images with real infrared images in terms of semantic and structural characteristics, we propose an IIAM, which adopts a joint spatial–frequency training strategy based on pretrained visible-domain weights to adapt latent reconstruction toward the infrared distribution, as illustrated in Fig. 3.

Specifically, IIAM is built upon a VAE with a fixed encoder and a trainable decoder. Given the real infrared image $\boldsymbol{X}^{ir}$ and the reconstructed infrared image $\hat{\boldsymbol{X}}^{ir}$, spatial-domain supervision is imposed by introducing an $\mathcal{L}_1$ loss to constrain their overall consistency, while an additional $\mathcal{L}_{\text{lpips}}$ is applied in both shallow and deep feature spaces to further enforce perceptual consistency. Furthermore, the real and reconstructed infrared images are transformed into the frequency domain [70], and their corresponding 2-D Fourier spectra are denoted as $\boldsymbol{F}_{\boldsymbol{X}^{ir}}(u,v)$ and $\hat{\boldsymbol{F}}_{\hat{\boldsymbol{X}}^{ir}}(u,v)$

$$\boldsymbol{F}_{\boldsymbol{X}^{ir}}(u,v) = \boldsymbol{R}_{\boldsymbol{X}^{ir}}(u,v) + j\,\boldsymbol{I}_{\boldsymbol{X}^{ir}}(u,v) \tag{1}$$

$$\hat{\boldsymbol{F}}_{\hat{\boldsymbol{X}}^{ir}}(u,v) = \hat{\boldsymbol{R}}_{\hat{\boldsymbol{X}}^{ir}}(u,v) + j\,\hat{\boldsymbol{I}}_{\hat{\boldsymbol{X}}^{ir}}(u,v) \tag{2}$$

where $j^2 = -1$, $\boldsymbol{R}_{\boldsymbol{X}^{ir}}$ and $\boldsymbol{I}_{\boldsymbol{X}^{ir}}$ denote the real and imaginary parts of the real-image spectrum, and $\hat{\boldsymbol{R}}_{\hat{\boldsymbol{X}}^{ir}}$ and $\hat{\boldsymbol{I}}_{\hat{\boldsymbol{X}}^{ir}}$ denote those of the reconstructed-image spectrum.

To reduce the frequency-domain discrepancy and emphasize difficult to optimize components, a weighting matrix is constructed as

$$\boldsymbol{W}(u,v) = \Big(\big|\boldsymbol{R}_{\boldsymbol{X}^{ir}}(u,v) - \hat{\boldsymbol{R}}_{\hat{\boldsymbol{X}}^{ir}}(u,v)\big| + \big|\boldsymbol{I}_{\boldsymbol{X}^{ir}}(u,v) - \hat{\boldsymbol{I}}_{\hat{\boldsymbol{X}}^{ir}}(u,v)\big|\Big)^{\gamma} \tag{3}$$

where γ is a scaling factor, which is set to 2 in our experiments.

Then, the frequency-domain loss is defined as

$$\mathcal{L}_{\text{freq}} = \frac{1}{HW}\sum_{u=1}^{H}\sum_{v=1}^{W}\boldsymbol{W}(u,v) \cdot \left\|\boldsymbol{F}_{\boldsymbol{X}^{ir}}(u,v) - \hat{\boldsymbol{F}}_{\hat{\boldsymbol{X}}^{ir}}(u,v)\right\|_2^2 \tag{4}$$

where H and W denote the height and width of the frequency spectrum, respectively.

Finally, the overall training objective of IIAM is formulated as

$$\mathcal{L}_{\text{IIAM}} = \mathcal{L}_1 + \lambda_1\mathcal{L}_{\text{lpips}} + \lambda_2\mathcal{L}_{\text{freq}} \tag{5}$$

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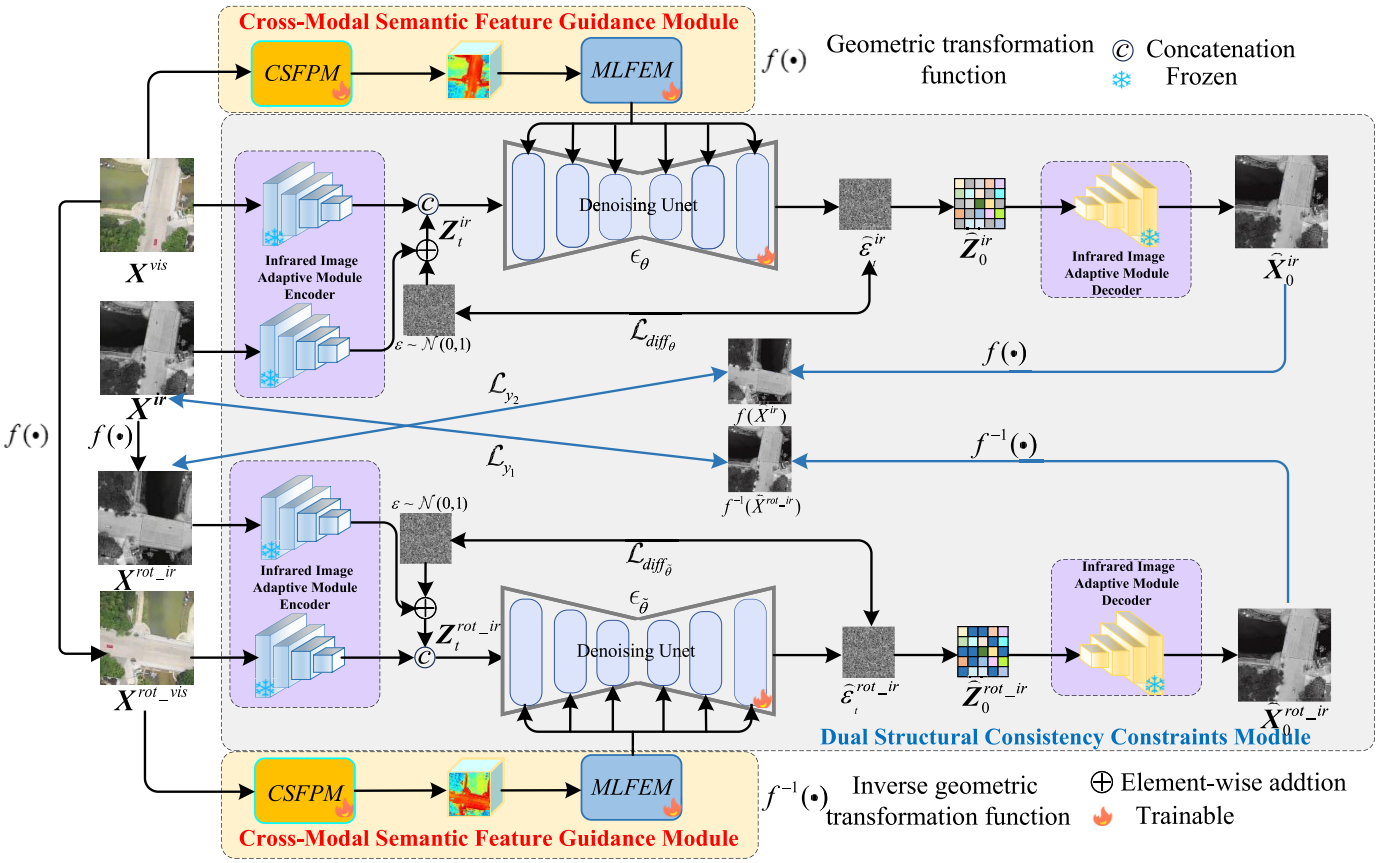


Fig. 2. Overall architecture of the proposed SSDiffNet. The framework consists of an IIAM, a CSFGM, and a DSCCM.

where λ_1 and λ_2 are weighting coefficients used to balance $\mathcal{L}_{\text{lpips}}$ and $\mathcal{L}_{\text{freq}}$. From the perspective of aerial infrared imaging, image fidelity depends not only on preserving global thermal radiation distribution but also on maintaining scene structure and local thermal details. Therefore, the proposed objective combines spatial domain, learned perceptual image patch similarity (LPIPS), and frequency-domain losses to provide complementary constraints on radiometric consistency, perceptual consistency, and multiscale spectral information, enabling the decoder to better capture the intrinsic characteristics of aerial infrared imagery.

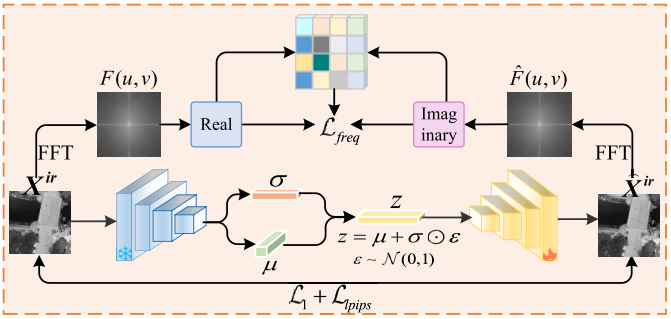


Fig. 3. Illustration of the training process of the IIAM.

C. Cross-Modal Semantic Feature Guidance Module

To enforce semantic consistency between generated and real infrared images, we propose a CSFGM inspired by UNIV [71]. The module consists of a cross-modal semantic feature prior module (CSFPM) and a multilevel feature embedding module (MLFEM). The CSFPM models cross-modal semantic correspondence through a dual-branch hybrid convolution Transformer architecture with contrastive learning, and the MLFEM injects the learned semantic priors into the generation process via spatial interactions.

1) Cross-Modal Semantic Feature Prior Module: The CSFPM adopts a dual-branch architecture during training, consisting of two structurally identical branches, denoted as $\boldsymbol{B}_{\theta_1}$ and $\boldsymbol{B}_{\theta_2}$. Each branch is composed of a three-stage encoder, including hierarchical masked convolution layers and a Transformer module. Specifically, the first two stages employ masked convolution to encode the input image into feature maps $\boldsymbol{F}_1 \in \mathbb{R}^{H/4\times W/4\times C_1}$ and $\boldsymbol{F}_2 \in \mathbb{R}^{H/8\times W/8\times C_2}$, which capture local structural patterns and fine-grained details. Local inductive biases are introduced at the early encoding stages to enhance the modeling of spatial layout information. The third stage introduces a Transformer encoder to map $\boldsymbol{F}_2$ into $\boldsymbol{F}_3 \in \mathbb{R}^{H/16\times W/16\times C_3}$, enabling global context modeling and long-range dependency capture. The overall architecture and training process are illustrated in Fig. 4.

To characterize the semantic relationships among patches within visible images under the unsupervised setting, we introduce an attention-based binary matrix construction strategy to provide structured supervision for cross-modal semantic

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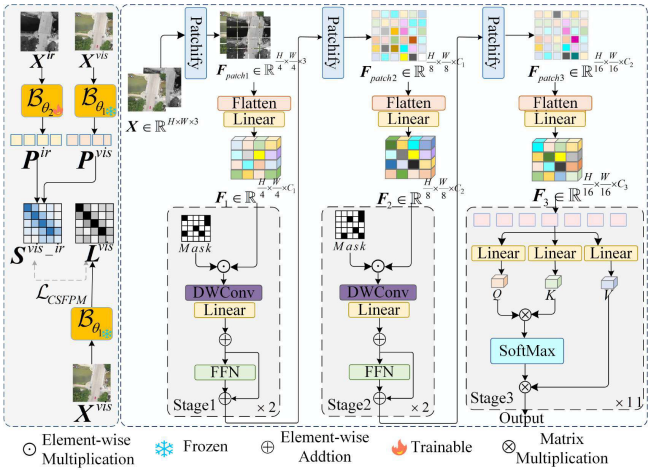


Fig. 4. Illustration of the training process and structural details of the CSFPM. The left side shows the training process, and the right side shows the network structure.

Algorithm 1 Binary Matrix Generation From Attention Maps

Require: Last-stage self-attention map $\boldsymbol{S}^{\text{vis_vis}} \in \mathbb{R}^{N\times N}$, threshold τ
Ensure: Binary pseudo-label matrix $\boldsymbol{L}^{\text{vis}}$
Initialize $\boldsymbol{L}^{\text{vis}} \leftarrow \boldsymbol{0}$
for $i = 1, \ldots, N$ **do**
  $\boldsymbol{L}^{\text{vis}}_{ii} \leftarrow 1$
  $\{\boldsymbol{S}^{\text{vis_vis}}_{ik}\}_{k=1}^{N} \leftarrow \boldsymbol{S}^{\text{vis_vis}}(i,:)$
  $\pi(\cdot) \leftarrow \text{SortIndicesDescending}(\{\boldsymbol{S}^{\text{vis_vis}}_{ik}\}_{k=1}^{N})$
$n_{\text{thr}} \leftarrow \min\left\{n \mid \sum_{k=1}^{n} \boldsymbol{S}^{\text{vis_vis}}_{i\pi(k)} > \tau\right\}$
  for $m = 1, \ldots, n_{\text{thr}}$ **do**
    $\boldsymbol{L}^{\text{vis}}_{i\pi(m)} \leftarrow 1$
  end for
end for
return $\boldsymbol{L}^{\text{vis}}$

alignment. Specifically, during the training of the CSFPM, branch $\boldsymbol{B}_{\theta_1}$ is kept frozen and loaded with weights pretrained on visible images [72], acting as a fixed visible-domain encoder. The self-attention feature map $\boldsymbol{S}^{vis_vis}$ is extracted from the final Transformer layer of $\boldsymbol{B}_{\theta_1}$

$$\boldsymbol{S}^{vis_vis} = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{\top}}{\sqrt{d_k}}\right) \in \mathbb{R}^{N\times N} \tag{6}$$

where $\boldsymbol{Q}$ and $\boldsymbol{K}$ denote the query and key matrices, respectively; d_k is the key dimension; N is the number of tokens; and softmax is applied row-wise. The map $\boldsymbol{S}^{vis_vis}$ reflects pairwise dependencies among image regions in the high-level semantic space. Since the original attention map is continuous and therefore difficult to use directly as a stable supervisory signal, we transform it into a binary matrix $\boldsymbol{L}^{vis}$. The detailed generation procedure of $\boldsymbol{L}^{vis}$ is presented in Algorithm 1.

Subsequently, we adopt a parameter-efficient fine-tuning strategy based on LoRA to optimize branch $\boldsymbol{B}_{\theta_2}$. The overall training process is formulated as a dual-branch semantic mapping framework, in which visible and infrared semantic features are extracted and aligned under the guidance of $\boldsymbol{L}^{vis}$. The detailed optimization pipeline is summarized in

Algorithm 2 CSFPM Training

Require: Visible images $\boldsymbol{X}^{vis}$, infrared images $\boldsymbol{X}^{ir}$, binary prior $\boldsymbol{L}^{vis}$
Ensure: Trained infrared semantic branch $\boldsymbol{B}_{\theta_2}$
Initialize $\boldsymbol{B}_{\theta_1}$ and $\boldsymbol{B}_{\theta_2}$ with pretrained weights
Freeze $\boldsymbol{B}_{\theta_1}$ and insert LoRA adapters into $\boldsymbol{B}_{\theta_2}$
repeat
  Extract patch features:

$$\boldsymbol{P}^{vis} = \boldsymbol{B}_{\theta_1}(\boldsymbol{X}^{vis}), \quad \boldsymbol{P}^{ir} = \boldsymbol{B}_{\theta_2}(\boldsymbol{X}^{ir})$$

  Compute cross-modal similarity:

$$\boldsymbol{S}^{vis_ir} = \text{Norm}(\boldsymbol{P}^{vis})\text{Norm}(\boldsymbol{P}^{ir})^{\top}$$

  Construct infrared-side prior:

$$\boldsymbol{L}^{ir} = (\boldsymbol{L}^{vis})^{\top}$$

  Optimize LoRA adapters with:

$$\mathcal{L}_{\text{CSFPM}} = \mathcal{L}_{\text{bce}}(\sigma(\boldsymbol{S}^{vis_ir}), \boldsymbol{L}^{vis}) + \mathcal{L}_{\text{bce}}(\sigma((\boldsymbol{S}^{vis_ir})^{\top}), \boldsymbol{L}^{ir})$$

  Update $\boldsymbol{B}_{\theta_2}$ through LoRA adapters
until Converged
return $\boldsymbol{B}_{\theta_2}$

Algorithm 2, where $\sigma(\cdot)$ denotes the sigmoid function. Norm$(\cdot)$ denotes row-wise ℓ_2 normalization of the patch features. Under this formulation, a bidirectional semantic alignment constraint is imposed through contrastive learning, in which cross-modal semantic features are jointly optimized from both infrared-to-visible and visible-to-infrared perspectives. $\mathcal{L}_{\text{bce}}$ encourages semantically consistent regions to be pulled closer while pushing apart semantically irrelevant ones, enabling the CSFPM to effectively capture the intrinsic correspondence between the two modalities. After training, $\boldsymbol{B}_{\theta_2}$ is able to learn a shared semantic representation across modalities, and the visible-to-infrared semantic prior $\boldsymbol{P}^{vis_ir}$ is extracted through the trained $\boldsymbol{B}_{\theta_2}$. Because the backbone of $\boldsymbol{B}_{\theta_2}$ retains its visible-domain initialization, while only its LoRA adapters are optimized for cross-modal alignment, $\boldsymbol{B}_{\theta_2}$ is applied to $\boldsymbol{X}^{vis}$ during generation to encode the visible input in the aligned representation space. Finally, only the trained branch $\boldsymbol{B}_{\theta_2}$ is preserved for visible-to-infrared generation in DSCCM.

2) Multilevel Feature Embedding Module: To effectively guide and modulate the internal representations of the generation network using the semantic information extracted by the CSFPM, we introduce an MLFEM. Specifically, during the visible-to-infrared generation process, the MLFEM is applied to the encoder layers, intermediate layers, and decoder layers of the U-Net backbone. Let the noise feature map at the ith layer of the U-Net be denoted as $\boldsymbol{U}_i$, which is treated as the query. Then, the cross-modal semantic prior features are used as keys and values. The cross-attention mechanism is formulated as follows:

$$\boldsymbol{Q}_i = \boldsymbol{W}_Q^{(i)} \cdot \boldsymbol{U}_i \tag{7}$$

$$\boldsymbol{K}_i = \boldsymbol{W}_K^{(i)} \cdot \boldsymbol{P}^{vis_ir} \quad \boldsymbol{V}_i = \boldsymbol{W}_V^{(i)} \cdot \boldsymbol{P}^{vis_ir} \tag{8}$$

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$$\boldsymbol{U}_i^{out} = \boldsymbol{U}_i + \text{softmax}\left(\frac{\boldsymbol{Q}_i \cdot \boldsymbol{K}_i^{\top}}{\sqrt{d}}\right) \cdot \boldsymbol{V}_i \tag{9}$$

where d denotes the feature dimension of the query and key embeddings.

D. Dual Structural Consistency Constraint Module

1) Structure of DSCCM: Considering that simple image rotation operations do not alter the semantic content of images [73], we design DSCCM, which integrates semantic priors from CSFGM and generates infrared images with more realistic structural and semantic properties via IIAM. First, the DSCCM consists of two branches with shared weights, where each branch is implemented by a diffusion model. Given the cross-modal prior features extracted by the trained $\boldsymbol{B}_{\theta_2}$, $\boldsymbol{P}^{vis_ir} = \boldsymbol{B}_{\theta_2}(\boldsymbol{X}^{vis})$, the forward diffusion process can be modeled as a Markov chain

$$\boldsymbol{Z}_0^{ir} = \boldsymbol{E}\left(\boldsymbol{X}^{ir}\right), \quad \boldsymbol{C}^{vis} = \boldsymbol{E}\left(\boldsymbol{X}^{vis}\right) \tag{10}$$

$$\boldsymbol{Z}_t^{ir} = \sqrt{\alpha_t}\,\boldsymbol{Z}_{t-1}^{ir} + \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon} \tag{11}$$

$$q\left(\boldsymbol{Z}_t^{ir} \mid \boldsymbol{Z}_{t-1}^{ir}\right) = \mathcal{N}\left(\boldsymbol{Z}_t^{ir} \mid \sqrt{\alpha_t}\,\boldsymbol{Z}_{t-1}^{ir},\ (1-\alpha_t)\,\boldsymbol{I}_{\text{lat}}\right) \tag{12}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \boldsymbol{I}_{\text{lat}})$, $\boldsymbol{I}_{\text{lat}}$ is the identity matrix in latent space, and α_t denotes the noise scheduling coefficient. Let the encoder and decoder of IIAM be denoted as $\boldsymbol{E}$ and $\boldsymbol{D}$, respectively. The first branch network ϵ_θ is employed to predict the noise in the latent space, and the predicted clean image at time step 0 can be represented as

$$\hat{\boldsymbol{X}}_0^{ir} = \boldsymbol{D}\left(\frac{1}{\sqrt{\bar{\alpha}_t}}\left(\boldsymbol{Z}_t^{ir} - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta\left(\boldsymbol{Z}_t^{ir}, t, \boldsymbol{C}^{vis}, \boldsymbol{P}^{vis_ir}\right)\right)\right) \tag{13}$$

where $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$ denotes the cumulative product of the noise scheduling coefficients. For the rotated branch $\epsilon_{\tilde{\theta}}$, where $\tilde{\theta} = \theta$ because the two branches share weights, a 90° rotation operation $f(\cdot)$ is applied to the visible image to obtain $\boldsymbol{X}^{rot_vis}$. The corresponding semantic guidance $\boldsymbol{P}^{rot_vis_ir} = \boldsymbol{B}_{\theta_2}(\boldsymbol{X}^{rot_vis})$ is then extracted by the CSFGM, and the paired rotated infrared image $\boldsymbol{X}^{rot_ir}$ is used as the reconstruction target. Similar to the original branch, we obtain $\boldsymbol{Z}_0^{rot_ir} = \boldsymbol{E}(\boldsymbol{X}^{rot_ir})$ and $\boldsymbol{C}^{rot_vis} = \boldsymbol{E}(\boldsymbol{X}^{rot_vis})$ and then sample $\boldsymbol{Z}_t^{rot_ir}$ through the forward diffusion process. After that, $\epsilon_{\tilde{\theta}}$ predicts the noise and the decoder of IIAM reconstructs the rotated initial prediction $\hat{\boldsymbol{X}}_0^{rot_ir}$. During inference, the rotated branch is discarded and only the original branch ϵ_θ is retained. To enhance the controllability of cross-modal semantic guidance, we adopt the classifier-free guidance (CFG) mechanism, where S_{vis} controls the contribution of the visible condition and S_{csf} controls the contribution of the cross-modal semantic features. The final noise prediction is composed of unconditional, visible conditioned, and semantic-guided predictions with predefined weights, formulated as follows:

$$\begin{aligned}\hat{\boldsymbol{\epsilon}}_\theta\left(\boldsymbol{Z}_t^{ir}, t, \boldsymbol{C}^{vis}, \boldsymbol{P}^{vis_ir}\right) &= \epsilon_\theta\left(\boldsymbol{Z}_t^{ir}, t, \varnothing, \varnothing\right) \\ &+ S_{vis} \cdot \left(\epsilon_\theta\left(\boldsymbol{Z}_t^{ir}, t, \boldsymbol{C}^{vis}, \varnothing\right) - \epsilon_\theta\left(\boldsymbol{Z}_t^{ir}, t, \varnothing, \varnothing\right)\right) \\ &+ S_{csf} \cdot \left(\epsilon_\theta\left(\boldsymbol{Z}_t^{ir}, t, \boldsymbol{C}^{vis}, \boldsymbol{P}^{vis_ir}\right) - \epsilon_\theta\left(\boldsymbol{Z}_t^{ir}, t, \boldsymbol{C}^{vis}, \varnothing\right)\right).\end{aligned} \tag{14}$$

2) Loss Function of DSCCM: First, the standard DDPM loss is used to train both the original and rotated branches

$$\mathcal{L}_{\text{diff}_\theta} = \mathbb{E}_{\boldsymbol{Z}_t^{ir}, \boldsymbol{\epsilon}, t, \boldsymbol{C}^{vis}, \boldsymbol{P}^{vis_ir}} \times \left[\left\|\boldsymbol{\epsilon} - \epsilon_\theta\left(\boldsymbol{Z}_t^{ir}, t, \boldsymbol{C}^{vis}, \boldsymbol{P}^{vis_ir}\right)\right\|_2^2\right] \tag{15}$$

$$\mathcal{L}_{\text{diff}_{\tilde{\theta}}} = \mathbb{E}_{\boldsymbol{Z}_t^{rot_ir}, \boldsymbol{\epsilon}, t, \boldsymbol{C}^{rot_vis}, \boldsymbol{P}^{rot_vis_ir}} \times \left[\left\|\boldsymbol{\epsilon} - \epsilon_{\tilde{\theta}}\left(\boldsymbol{Z}_t^{rot_ir}, t, \boldsymbol{C}^{rot_vis}, \boldsymbol{P}^{rot_vis_ir}\right)\right\|_2^2\right]. \tag{16}$$

Subsequently, an inverse rotation $f^{-1}(\cdot)$ is applied to $\hat{\boldsymbol{X}}_0^{rot_ir}$, while $f(\cdot)$ is applied to $\hat{\boldsymbol{X}}_0^{ir}$

$$\boldsymbol{Y}_1 = f^{-1}\left(\hat{\boldsymbol{X}}_0^{rot_ir}\right), \quad \boldsymbol{Y}_2 = f\left(\hat{\boldsymbol{X}}_0^{ir}\right). \tag{17}$$

$\mathcal{L}_1$ is then computed between $\boldsymbol{Y}_1$ and the real infrared image $\boldsymbol{X}^{ir}$, and between $\boldsymbol{Y}_2$ and the real rotated infrared image $\boldsymbol{X}^{rot_ir}$

$$\mathcal{L}_{y_1} = \left\|\boldsymbol{Y}_1 - \boldsymbol{X}^{ir}\right\|_1, \quad \mathcal{L}_{y_2} = \left\|\boldsymbol{Y}_2 - \boldsymbol{X}^{rot_ir}\right\|_1. \tag{18}$$

The dual geometric consistency loss is defined as

$$\mathcal{L}_{\text{dual}} = \left(1 - \frac{t}{T}\right)^2 \left(\mathcal{L}_{y_1} + \mathcal{L}_{y_2}\right) \tag{19}$$

where t denotes the current diffusion time step and T is the total number of diffusion steps.

Finally, the overall loss function of DSCCM is formulated as

$$\mathcal{L}_{\text{DSCCM}} = \mathcal{L}_{\text{diff}_\theta} + \lambda_3 \mathcal{L}_{\text{diff}_{\tilde{\theta}}} + \lambda_4 \mathcal{L}_{\text{dual}} \tag{20}$$

where λ_3 and λ_4 are hyperparameters used to balance the diffusion loss of the rotated branch and the dual geometric consistency loss, respectively.

IV. Experiments and Analysis

A. Datasets

The experiments in this article are conducted on the AVIID dataset [2] and the DroneVehicle dataset [74]. According to the acquisition time, the DroneVehicle dataset is further divided into the Day-DroneVehicle dataset and the Night-DroneVehicle dataset. The details of these three datasets are described as follows.

1) AVIID: AVIID is a dedicated dataset constructed by our team for the AVIT task. In this article, we use an updated version of AVIID that employs a more accurate scheme for registering visible and infrared images, effectively reducing interference from spatial misalignment. In the experiments, 2000 paired image samples are selected from this dataset, with a spatial resolution of 512 × 512. The scene types include roads, bridges, streets, and parking areas. The dataset is randomly divided into training and testing sets at an 80:20 ratio.

2) Day-DroneVehicle: The Day-DroneVehicle dataset is collected by unmanned aerial vehicles under daytime conditions and contains various typical urban scenes. In this article, 4000 pairs of visible and infrared images are selected as experimental samples from this dataset. The scenes cover urban roads, residential areas, industrial regions, and parking lots. All images are standardized to a spatial resolution of 640 × 512. The target vehicle categories include five types, namely cars, freight cars, trucks, buses, and vans. During the experiments, the dataset is randomly divided into training and testing sets at an 80:20 ratio.

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TABLE I

QUANTITATIVE COMPARISON ON THE AVIID, DAY-DRONEVEHICLE, AND NIGHT-DRONEVEHICLE DATASETS (↑ INDICATES THAT HIGHER VALUES ARE BETTER, AND ↓ INDICATES THAT LOWER VALUES ARE BETTER)

Methods	AVIID				Day-DroneVehicle				Night-DroneVehicle			
	PSNR↑	**SSIM↑**	**LPIPS↓**	**RMSE↓**	**PSNR↑**	**SSIM↑**	**LPIPS↓**	**RMSE↓**	**PSNR↑**	**SSIM↑**	**LPIPS↓**	**RMSE↓**
BBDM [29] (*CVPR 2023*)	22.101	0.621	0.204	17.221	12.595	0.295	0.424	64.087	16.682	0.509	0.263	48.022
InstructPix2Pix [61] (*CVPR 2023*)	23.361	0.671	0.102	17.201	16.048	0.521	0.169	40.090	17.004	0.555	0.220	35.610
DR-AVIT [67] (*TGRS 2024*)	13.725	0.360	0.351	51.737	12.644	0.442	0.233	62.612	12.899	0.306	0.344	59.863
StegoGAN [78] (*CVPR 2024*)	17.712	0.473	0.219	34.711	12.692	0.376	0.322	60.522	13.875	0.364	0.368	52.668
USTNet [17] (*TGRS 2025*)	20.445	0.579	0.201	24.808	14.544	0.478	0.224	50.231	14.367	0.437	0.301	49.611
PID [15] (*Pattern Recognition 2026*)	21.523	0.577	0.169	23.082	14.254	0.341	0.318	53.396	15.277	0.396	0.309	46.227
EditAR [79] (*CVPR 2025*)	14.541	0.323	0.377	60.663	11.926	0.166	0.573	63.343	13.209	0.231	0.558	62.212
Ours	**24.212**	**0.701**	**0.078**	**17.132**	**17.266**	**0.554**	**0.151**	**39.246**	**18.275**	**0.566**	**0.169**	**34.001**

3) Night-DroneVehicle: The Night-DroneVehicle dataset is mainly designed for cross-modal perception tasks under nighttime and extremely low illumination conditions, where visible images are significantly affected by complex lighting variations. In this article, 9000 pairs of visible and infrared image samples are selected from this dataset. The same data partition strategy as that used for Day-DroneVehicle is adopted, with 80% for training and 20% for testing.

B. Implementation Details

During the training of the IIAM, the Adam optimizer is used for parameter updates. The learning rate is set to 1×10^{-4}, the batch size is set to 4, and the number of training epochs is 20. For the IIAM objective, the weight of the LPIPS loss λ_1 is set to 0.1, and the weight of the frequency-domain loss λ_2 is set to 0.01. During training, the encoder parameters are frozen to preserve the encoding capability for visible images. The parameters of the CSFPM are initialized from an ImageNet-1K-pretrained MCMAE model [72], and the model is trained for 400 epochs to ensure sufficient convergence. In this article, the threshold τ is set to 0.6. In the final training stage of the DSCCM, the diffusion loss weight of the rotated branch λ_3 is set to 0.01, and the dual geometric consistency loss weight λ_4 is also set to 0.01. The weights of the four loss terms, λ_1–λ_4, are empirically determined by considering the relative contributions and numerical scales of the corresponding loss terms. We set $\lambda_1 = 0.1$ and $\lambda_2 = \lambda_3 = \lambda_4 = 0.01$ throughout all experiments. The same configuration is adopted across the three datasets to maintain a consistent optimization objective and avoid dataset-specific parameter tuning. All images are first resized to 288×288 and then cropped to 256×256 to improve both training and inference efficiency. The model is trained for 300 epochs to ensure sufficient convergence. All experiments are implemented in PyTorch and conducted on four NVIDIA A100 GPUs, each with 40 GB of memory.

C. Evaluation Metrics and Comparison Methods

To quantitatively evaluate the quality of the generated infrared images from visible inputs, four commonly used evaluation metrics are adopted in this article, including peak signal-to-noise ratio (PSNR) [75], structural similarity index (SSIM) [76], LPIPS [77], and root-mean-square error (RMSE). These metrics characterize the discrepancy between the generated infrared images and the real infrared images from different perspectives, including pixel-level reconstruction accuracy, structural fidelity, high-level perceptual similarity, and overall pixel-wise error.

Using these criteria, we compare SSDiffNet with several recent representative methods, including BBDM [29], InstructPix2Pix [61], DR-AVIT [67], StegoGAN [78], USTNet [17], PID [15], and EditAR [79].

D. Results and Analysis

1) Quantitative Analysis: As shown in Table I, the overall quantitative evaluation results indicate that different methods exhibit clear performance differences across multiple datasets and imaging conditions.

From the perspective of pixel-level reconstruction metrics, BBDM [29] maintains relatively stable PSNR and SSIM values, indicating its advantage in modeling global brightness distributions and low-frequency structures. However, its LPIPS values are relatively high, suggesting that perceptual discrepancies persist between the generated and real infrared images in local texture representations. InstructPix2Pix [61] achieves relatively balanced performance on PSNR and SSIM and obtains lower LPIPS values than some diffusion-based methods, showing advantages in perceptual similarity. Although DR-AVIT [67] is specifically designed for aerial visible-to-infrared translation, it yields relatively low PSNR and SSIM values and higher LPIPS scores, indicating that its content-style disentanglement mechanism remains limited in structural preservation. StegoGAN [78] achieves moderate SSIM results, but its LPIPS values remain high, indicating that the generated results lack perceptual consistency without explicit structural and semantic constraints.

Among methods with stronger feature modeling capability, USTNet [17] shows advantages in terms of SSIM and LPIPS, demonstrating the potential of long-range dependency modeling for preserving structural consistency and perceptual similarity. However, its PSNR values are relatively low, indicating that pixel-level reconstruction accuracy remains limited. PID [15] introduces physical priors and achieves relatively balanced PSNR and SSIM results, but its LPIPS values remain higher than those of the best-performing methods, suggesting that noticeable deviations persist in complex scenes. EditAR [79] yields overall lower values across all metrics, indicating that direct distillation from large visible-domain models may introduce modality bias in cross-modal infrared generation tasks.

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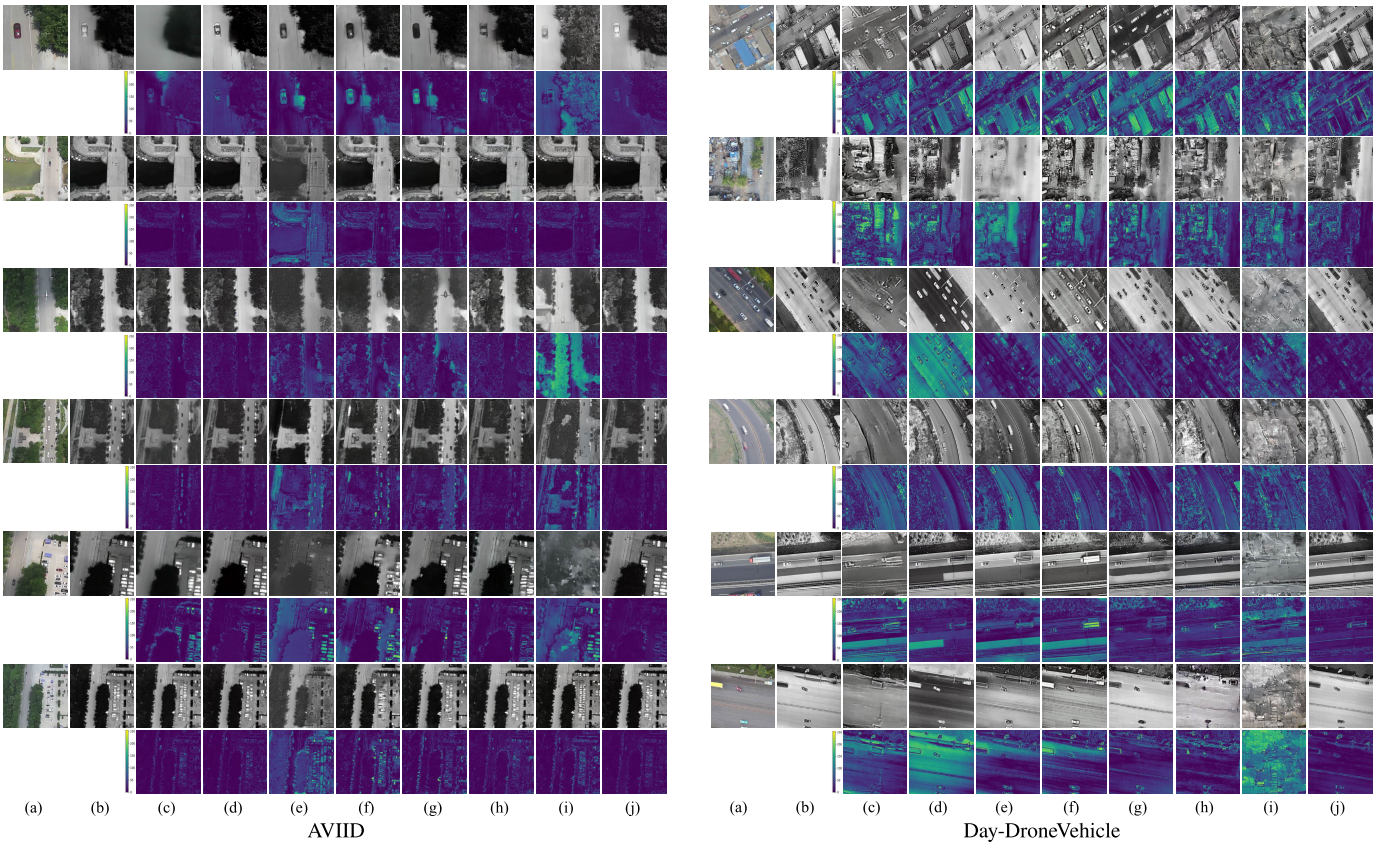


Fig. 5. Visual comparison results of different methods on the AVIID and Day-DroneVehicle datasets. (a) and (b) Visible image and real infrared image, respectively. (c)–(j) Results of BBDM, InstructPix2Pix, DR-AVIT, StegoGAN, USTNet, PID, EditAR, and our method. The residual maps are shown below the generated results of each method.

In contrast, our method achieves the best PSNR, SSIM, and LPIPS simultaneously. Relative to the best competing method on each dataset, SSDiffNet improves PSNR by 3.64%–7.59% and SSIM by 1.98%–6.33% while reducing LPIPS by 10.65%–23.53% and RMSE by 0.40%–4.52%. These results demonstrate more reliable modeling of global brightness distributions, spatial structures, and perceptual similarity across the three datasets.

Overall, by explicitly introducing cross-modal semantic guidance and imposing structural consistency constraints during the diffusion generation process, we effectively alleviate common issues such as structural drift and semantic inconsistency in cross-modal infrared image generation. As a result, our method achieves more stable, balanced, and reliable quantitative performance under different imaging conditions and complex aerial scenes.

To comprehensively assess the visual quality and semantic fidelity of the generated infrared images, downstream object detection experiments are conducted on the Day-DroneVehicle dataset using two representative detectors, YOLOv5-s and YOLOv26-s, with the results reported in Table II. YOLOv5-s is employed as a widely used benchmark for comparing different image translation methods, whereas YOLOv26-s is further adopted to evaluate the generalization and robustness of the proposed method across different detector architectures. Although real infrared images (GT) provide the upper bound detection performance, SSDiffNet achieves the best mAP and $F1$-scores among all generative methods with both detectors, together with competitive precision and recall. These results demonstrate that SSDiffNet more effectively preserves target structures and semantic information, thereby providing more reliable infrared images for downstream object detection.

TABLE II

QUANTITATIVE RESULTS OF VEHICLE DETECTION ON THE DAY-DRONEVEHICLE DATASET WITH YOLOV5-S AND YOLOV26-S DETECTORS (↑ INDICATES THAT HIGHER VALUES ARE BETTER)

Methods	YOLOv5-s				YOLOv26-s			
	P(%)↑	R(%)↑	mAP(%)↑	F1(%)↑	P(%)↑	R(%)↑	mAP(%)↑	F1(%)↑
GT	65.5	51.8	59.6	57.8	80.7	65.3	73.7	72.3
BBDM [29]	16.1	12.6	13.6	14.1	31.9	11.8	20.2	17.2
DR-AVIT [67]	54.0	41.4	48.3	46.8	**75.4**	36.5	55.6	49.6
StegoGAN [78]	54.9	36.5	45.9	43.9	70.4	30.1	49.6	42.2
USTNet [17]	22.1	44.2	18.6	29.5	71.4	**49.7**	60.4	58.4
PID [15]	23.8	12.3	16.7	16.2	38.0	8.6	22.5	14.0
Ours	**59.4**	**48.1**	**54.6**	**53.2**	74.1	48.3	**60.9**	**58.6**

2) Visual Comparison: We further conduct visual comparisons with other methods, as shown in Figs. 5–7.

The visual results reveal substantial differences among methods in the modeling of brightness semantics and the preservation of structural details for aerial visible-to-infrared generation. BBDM [29] produces relatively stable infrared images in terms of overall road structures and major target locations. However, the infrared response intensity of target regions is relatively weak, and the thermal contrast between targets and backgrounds is insufficient. The residual

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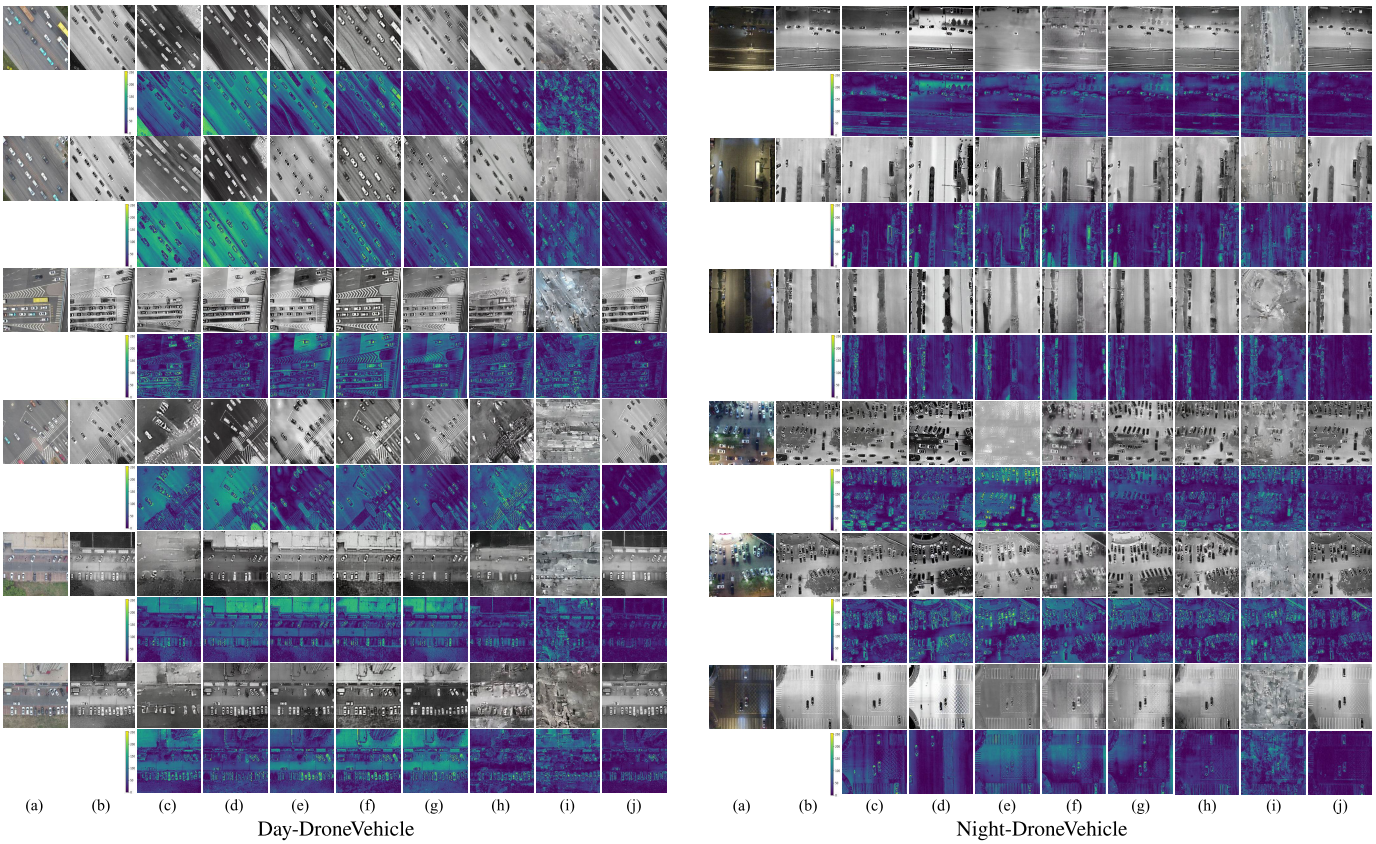


Fig. 6. Visual comparison results of different methods on the Day-DroneVehicle and Night-DroneVehicle datasets. (a) and (b) Visible image and real infrared image, respectively. (c)–(j) Results of BBDM, InstructPix2Pix, DR-AVIT, StegoGAN, USTNet, PID, EditAR, and our method. The residual maps are shown below the generated results of each method.

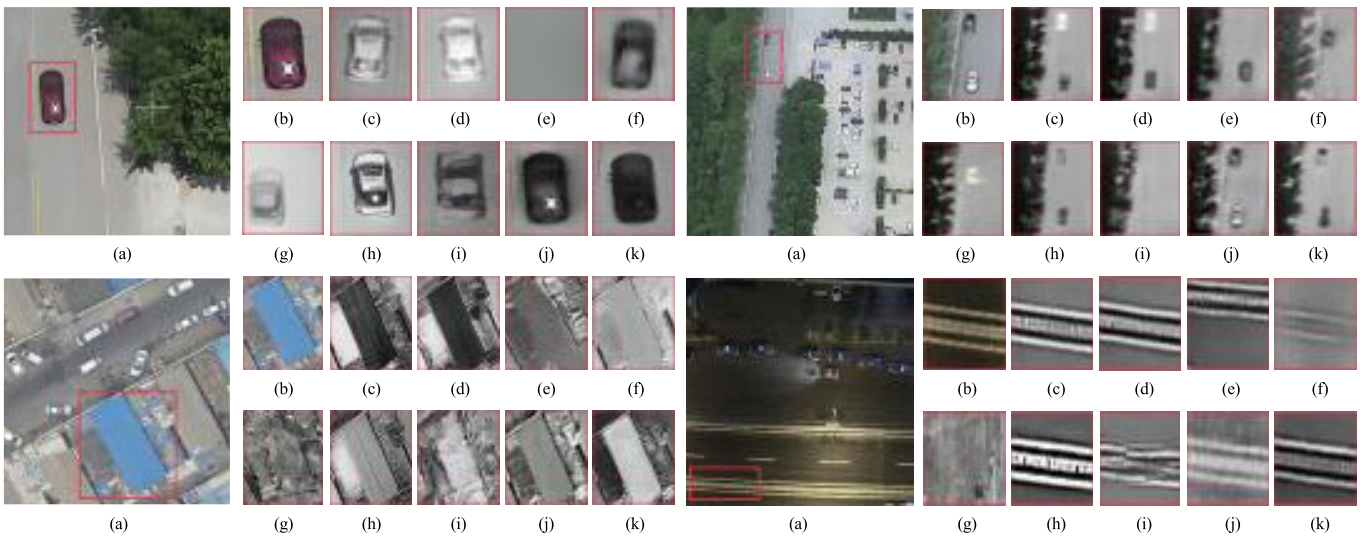


Fig. 7. Zoomed-in results of different methods on the AVIID, Day-DroneVehicle, and Night-DroneVehicle datasets. (a) Visible image, (b) zoomed-in region of the visible image, and (c) zoomed-in region of the real infrared image. (d)–(k) Zoomed-in results of our method, BBDM, DR-AVIT, EditAR, InstructPix2Pix, PID, StegoGAN, and USTNet, respectively.

maps still show noticeable high-error regions around vehicles and road boundaries. InstructPix2Pix [61] achieves global structural layouts relatively similar to those of real infrared images, but deviations remain in the modeling of brightness semantics. The thermal intensity distributions of some target regions are inconsistent, and the residual maps indicate error concentration in local areas. DR-AVIT [67] exhibits unstable brightness distributions in certain scenes, where local regions are overly smoothed. Fine-grained textures and target boundaries are insufficiently preserved, and the residual maps

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Fig. 8. Cross-dataset visual results on the Day-DroneVehicle and Night-DroneVehicle datasets. (a) Results trained on Night-DroneVehicle and tested on Day-DroneVehicle. (b) Real infrared images from Day-DroneVehicle. (c) Results trained on Day-DroneVehicle and tested on Night-DroneVehicle. (d) Real infrared images from Night-DroneVehicle.

show widespread errors. StegoGAN [78] maintains relatively complete structural contours, but brightness drift remains in detailed regions, especially around vehicle edges and road textures. Continuous high-error regions can be observed in the residual maps, indicating limitations in the joint modeling of fine-grained structures and brightness semantics. USTNet [17] shows improvements in local structural modeling, with clearer target contours in some cases. However, its overall brightness levels still differ from those of real infrared images, and brightness shifts frequently occur in background regions. The corresponding residual maps still contain scattered error regions. PID [15] demonstrates relatively balanced structural consistency, but inaccurate brightness responses remain in complex scenes. Obvious error distributions can be observed in both target regions and complex backgrounds in the residual maps. EditAR [79] relies on visible-domain prior distillation, which leads to large deviations in the modeling of infrared brightness semantics. The thermal distribution relationships between targets and backgrounds are distorted, and large-scale high-error regions appear in the residual maps.

In contrast, we generate infrared images that are much closer to real infrared images in terms of overall brightness distribution, target thermal responses, and local detail representation. The thermal contrast relationships among vehicles, roads, and backgrounds are more natural and stable. The corresponding residual maps show a significant reduction in high-error regions, indicating that our method can more accurately learn the brightness semantic correspondence between visible and infrared images while maintaining structural consistency.

Overall, the visual results across the three datasets demonstrate that our method achieves stronger semantic consistency and structural stability in various complex aerial scenarios.

To further assess the robustness and cross-dataset generalization capability of SSDiffNet, cross-dataset experiments

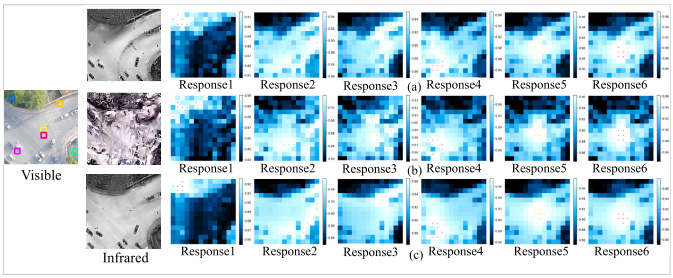


Fig. 9. Cross-modal semantic response visualization between the selected visible image patches and their corresponding patches in different infrared images. (a) Visible with real infrared. (b) Visible with PID-generated infrared. (c) Visible with infrared generated by our method. Stronger responses indicate higher cross-modal semantic consistency. Best viewed in color and zoomed in.

are conducted between the Day-DroneVehicle and Night-DroneVehicle datasets. Specifically, the model is trained on one dataset and directly evaluated on the other without additional fine-tuning. As illustrated in Fig. 8, SSDiffNet consistently produces infrared images with clearly delineated road structures, well-preserved vehicle regions, and realistic thermal characteristics, despite the substantial domain discrepancy between the two datasets. These results demonstrate that SSDiffNet exhibits strong robustness and generalization across diverse aerial infrared imaging conditions.

We further conduct a comparative semantic similarity analysis of the proposed method and a representative competing method to investigate their differences in cross-modal semantic correspondence, as shown in Fig. 9.

From the cross-modal semantic response visualizations, it can be observed that for the same visible image patch, the corresponding patch in the real infrared image exhibits relatively concentrated high response regions. This indicates a consistent semantic correspondence between the two modalities at the semantic level. In contrast, for the infrared images generated by the PID [15] method, the response distributions between visible patches and their corresponding infrared patches are more scattered. Response shifts and semantic confusion can be observed in some regions, which suggests that the generated results still suffer from limitations in cross-modal semantic alignment.

By comparison, the infrared images generated by our method show higher consistency with real infrared images in terms of cross-modal semantic response distributions. For each visible image patch, the high response regions in the generated infrared images are closer to those of the real infrared images in both spatial location and response patterns. This indicates that the generated images more stably and consistently preserve the semantic structures and contextual relationships of target regions. Therefore, our method can effectively maintain the semantic correspondence between visible and infrared images during the generation process, enabling the generated infrared images to exhibit semantic response patterns that are more similar to real infrared images at the local patch level. This further verifies the effectiveness of our method in modeling cross-modal semantic consistency.

We also visualize the results of the downstream detection task, as shown in Fig. 10. The detection results obtained

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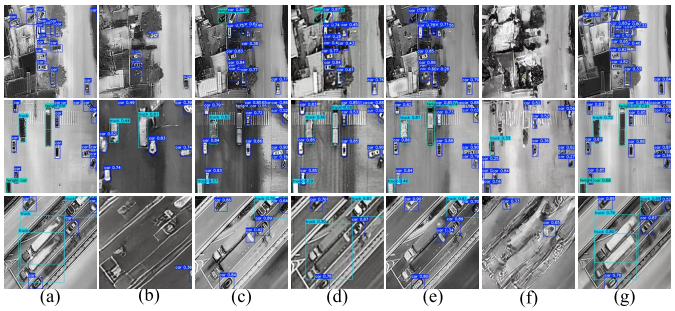


Fig. 10. Visual comparison of detection results obtained by different methods on the Day-DroneVehicle dataset. (a) Ground-truth annotations. (b)–(g) Results of BBDM, DR-AVIT, StegoGAN, USTNet, PID, and our method, respectively.

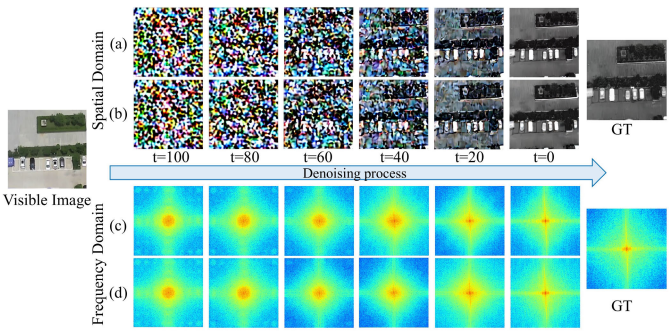


Fig. 11. Visualization of the progressive denoising process. (a) Spatial-domain denoising with IIAM. (b) Spatial-domain denoising without IIAM. (c) Frequency-domain denoising with IIAM. (d) Frequency-domain denoising without IIAM.

TABLE III

COMPARISON BETWEEN VISIBLE-DOMAIN VAE AND INFRARED-ADAPTED VAE ON THE AVIID DATASET

Methods	PSNR ↑	SSIM ↑	LPIPS ↓	RMSE ↓
Visible-domain VAE	23.342	0.669	0.094	17.730
Infrared-adapted VAE (IIAM)	**23.661**	**0.682**	**0.091**	**17.168**

from infrared images generated by different methods exhibit clear performance differences in downstream object detection tasks. Some methods are able to produce a relatively large number of detection boxes, but they generally suffer from frequent false positives, unstable bounding-box locations, and incomplete target structures. In contrast, the detection results of our method are the closest to the ground-truth annotations. The predicted bounding boxes show higher consistency in terms of number, spatial location, and scale distribution, while both false detections and missed detections are significantly reduced. This demonstrates that the generated infrared images are not only visually closer to real infrared data but also preserve more reliable target contours, local structures, and brightness semantics, thereby providing more effective support for downstream object detection tasks.

E. Ablation Experiments and Parameter Analysis

1) Ablation Experiments: To investigate the contribution of the proposed IIAM, we first compare it with a baseline VAE trained on visible images. The quantitative results are reported in Table III. Compared with the visible-domain VAE, IIAM consistently achieves superior performance on all evaluation metrics. Specifically, PSNR and SSIM improve by 1.37% and 1.94%, respectively, while LPIPS and RMSE decrease by 3.19% and 3.17%. These results demonstrate that adapting the decoder to the infrared domain enables the model to better capture infrared-specific appearance and radiometric distributions, thereby improving the quality of the generated infrared images.

Furthermore, we use 100 denoising steps for inference in both the baseline model and the improved model with IIAM. The generated results are visualized at 20-step intervals to analyze the spatial domain and frequency-domain evolution during denoising. This allows a direct comparison of the two models' behavior across different denoising stages, as shown in Fig. 11. After IIAM modulation, the frequency spectra of the generated infrared images become closer to those of real infrared images. This agreement is most evident in the middle and late denoising stages. Compared with the baseline, IIAM produces a more stable low-frequency center and a clearer medium- and high-frequency attenuation pattern. This shows that IIAM helps the model learn the brightness distribution and spectral structure of infrared images.

Overall, IIAM improves the consistency of spatial structure, frequency distribution, and infrared appearance.

TABLE IV

ABLATION STUDY RESULTS ON THE AVIID, DAY-DRONEVEHICLE, AND NIGHT-DRONEVEHICLE DATASETS

Dataset	Methods	PSNR ↑	SSIM ↑	LPIPS ↓	RMSE ↓
AVIID	Baseline	23.361	0.671	0.102	17.201
	+ IIAM	23.661	0.682	0.091	17.168
	+ CSFGM	24.035	0.694	0.083	17.149
	+ DSCCM	24.058	0.693	0.082	17.141
	Ours	**24.212**	**0.701**	**0.078**	**17.132**
Day	Baseline	16.048	0.521	0.169	40.090
	+ IIAM	16.477	0.533	0.161	39.686
	+ CSFGM	17.012	0.546	0.155	39.454
	+ DSCCM	17.045	0.545	0.154	39.356
	Ours	**17.266**	**0.554**	**0.151**	**39.246**
Night	Baseline	17.004	0.555	0.220	35.610
	+ IIAM	17.452	0.559	0.197	34.840
	+ CSFGM	18.011	0.563	0.180	34.397
	+ DSCCM	18.045	0.561	0.178	34.210
	Ours	**18.275**	**0.566**	**0.169**	**34.001**

We further conduct ablation experiments on three datasets to evaluate the effectiveness of the proposed modules. As reported in Table IV, each module consistently improves the baseline, and combining all modules achieves the best performance. Compared with the baseline, the complete model improves PSNR and SSIM by an average of 6.23% and 4.26%, respectively, while reducing LPIPS and RMSE by 19.12% and 2.34%. These results indicate that IIAM, CSFGM, and DSCCM provide complementary benefits for infrared domain adaptation, semantic guidance, and structural preservation.

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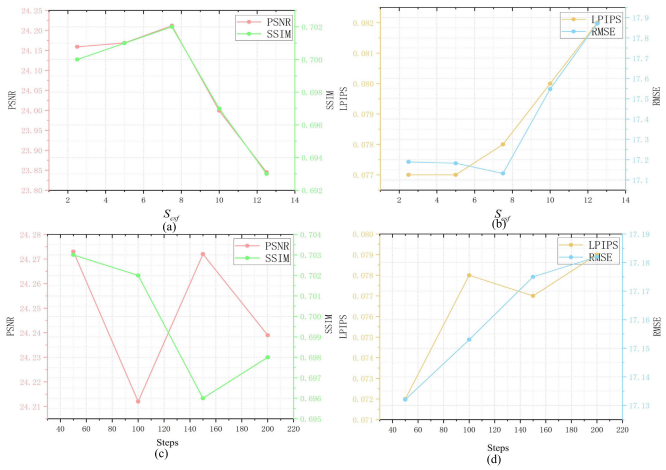


Fig. 12. Line plots of performance metrics under different parameter settings. (a) and (c) PSNR and SSIM under different settings of S_{csf} and the number of sampling steps. (b) and (d) LPIPS and RMSE under the same parameter variations.

TABLE V
QUANTITATIVE COMPARISON OF DIFFERENT GEOMETRIC TRANSFORMATIONS IN DSCCM

Transformation	PSNR ↑	SSIM ↑	LPIPS ↓	RMSE ↓
90°	**24.212**	**0.701**	**0.078**	**17.132**
180°	24.210	0.700	0.084	17.134
270°	24.161	**0.701**	**0.078**	17.191
Scaling	23.480	0.694	0.089	17.471
Flip	24.035	0.699	0.083	17.284
Affine	24.160	0.698	0.082	17.350

2) Parameter Analysis: In addition, we investigate the effects of the number of sampling steps and the cross-modal semantic control coefficient S_{csf}, as shown in Fig. 12.

We set the number of sampling steps to 100 to ensure a stable denoising trajectory while maintaining a good balance between generation quality and inference efficiency.

In addition, the cross-modal semantic control coefficient S_{csf} is used to adjust the strength of the semantic guidance network during the generation process. Experimental results indicate that increasing S_{csf} within a certain range helps improve the semantic consistency and structural realism of the generated images, which demonstrates that appropriate cross-modal semantic priors can enhance the guidance and control capability of the model. However, when the control coefficient becomes excessively large, overly strong semantic constraints may interfere with the original visible information, which further leads to a degradation in generation quality.

V. DISCUSSION

This section discusses the key experimental findings of SSDiffNet, examines its current limitations, and identifies several promising directions for future research.

Given that the effectiveness of geometric transformations is crucial to DSCCM, Table V provides a detailed comparison of different transformation strategies, including rotation, flipping, scaling, and affine transformation, as well as rotations at different angles. The results show that all considered transformations improve translation performance by introducing additional structural constraints during diffusion training. Among them, the fixed 90° rotation consistently achieves the best overall performance, thereby supporting the design choice adopted in DSCCM. One possible explanation is that aerial targets naturally exhibit diverse orientations, and a 90° rotation can effectively increase orientation diversity without changing object scale or introducing the geometric distortions and interpolation artifacts commonly associated with scaling and affine transformations. These findings indicate that geometry-preserving transformations are particularly effective for learning robust structural correspondences between visible and infrared images.

Despite these advantages, SSDiffNet still has several limitations. As illustrated in Fig. 7, although the proposed method generates realistic infrared images with well-preserved global structures, the contours of small targets may still appear weakened or blurred. This issue mainly arises because vehicles in aerial imagery often occupy only a limited number of pixels, making their boundaries particularly vulnerable to smoothing during the generation process. In addition, the current optimization objective focuses primarily on global scene consistency and overall infrared appearance, without explicitly constraining local object structures. Future work could therefore introduce target-aware supervision, such as detection-guided objectives or object structure preservation losses. Increasing the input resolution, for example, to 512×512, and incorporating additional structural priors, such as vehicle annotations or edge information, may further improve the preservation of fine-grained target contours.

In addition to improving spatial detail preservation, extending SSDiffNet to visible-to-infrared video translation represents another promising research direction. A straightforward solution would be to apply the current framework independently to consecutive video frames. However, frame-wise translation may lead to temporal inconsistencies, such as fluctuations in brightness, structural details, and target responses across adjacent frames. Future studies could incorporate temporal attention mechanisms, motion-aware feature propagation, or interframe consistency constraints to enhance temporal coherence while maintaining stable infrared appearance and target characteristics throughout the generated video sequence. Beyond video translation, extending SSDiffNet to other challenging and practically valuable cross-modal translation tasks, such as RGB-to-SAR image translation, is also worth exploring. This extension may require adapting the proposed components to visible and SAR data and further investigating SAR-specific constraints.

VI. CONCLUSION

In this article, we presented SSDiffNet, a diffusion-based framework for aerial visible-to-infrared image translation. By jointly incorporating infrared image adaptation, cross-modal semantic feature guidance, and dual structural consistency constraints, SSDiffNet narrows the modality gap while preserving critical semantic content and structural details. Experiments

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on multiple aerial datasets show that SSDiffNet consistently surpasses existing methods in terms of both visual quality and quantitative performance. The generated images also perform competitively in downstream aerial object detection, supporting their use for infrared data augmentation and cross-modal aerial perception. Future work will focus on improving inference efficiency through reduced diffusion steps, more efficient sampling solvers, and distillation-based acceleration strategies. We will also investigate stronger mechanisms for preserving fine-grained targets and extend SSDiffNet to aerial visible-to-infrared video generation.

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